

Exact Fourier Blocks and the Essential-Norm Boundary of the Circular Telegraph Semigroup

Internal mathematical artifact – HCS-C213

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Abstract

We give a complete source-local theorem for the velocity-jump process on the circle. A speed $c \geq 0$ transports a sign velocity and a rate $\lambda \geq 0$ flips it. Every Fourier mode is a two-by-two block whose centered square is scalar. This yields an exact matrix exponential, the telegraph equation, a sharp spectral-abscissa gap, all stationary boundaries, and the essential-norm obstruction. Critical Jordan blocks and both degenerate faces are included. The finite receipt is a regression witness, not a proof by enumeration. Route A is rejected under the declared scope.

Revision status. Revision checkpoint: essential-norm proof and all degenerate stationary faces closed. AI-use disclosure: language assistance was used for drafting and code review; all displayed identities and receipts are independently checked.

1 Owner and block reduction

Let $\mathbb{T}_{2\pi} = \mathbb{R}/(2\pi\mathbb{Z})$ and let $v \in \{+1, -1\}$. On normalized $L^2(\mathbb{T}_{2\pi} \times \{\pm 1\})$ consider

$$Lf(x, v) = cv \partial_x f(x, v) + \lambda(f(x, -v) - f(x, v)).$$

Here L is the backward (observable) generator with periodic domain $D(L) = H_{\text{per}}^1(\mathbb{T}_{2\pi}) \otimes \mathbb{C}^2$. Densities evolve under the adjoint L^* ; its Fourier blocks are the conjugate-index blocks G_{-k} , so the spectra and block norms below are unchanged. The clock is physical elapsed time. For the Fourier coefficient of mode $k \in \mathbb{Z}$ in the ordered velocity basis,

$$G_k = \begin{pmatrix} -\lambda + ick & \lambda \\ \lambda & -\lambda - ick \end{pmatrix}, \quad N_k = G_k + \lambda I.$$

Direct multiplication gives

$$N_k^2 = (\lambda^2 - c^2 k^2)I. \tag{1}$$

2 Exact semigroup and critical blocks

Put $\delta_k^2 = \lambda^2 - c^2 k^2$. Equation (1) and the power series for the exponential imply

$$e^{tG_k} = e^{-\lambda t} \left(\cosh(\delta_k t) I + \frac{\sinh(\delta_k t)}{\delta_k} N_k \right), \tag{2}$$

where the quotient is t when $\delta_k = 0$. Thus hyperbolic, oscillatory, and critical regimes are one formula. The characteristic polynomial is

$$\det(rI - G_k) = (r + \lambda)^2 - (\lambda^2 - c^2 k^2), \quad \rho_{k,\pm} = -\lambda \pm \sqrt{\lambda^2 - c^2 k^2}.$$

If $\lambda = c|k| > 0$, the nonzero nilpotent N_k gives one Jordan block at $-\lambda$. At $k = 0$ the two eigenvalues are 0 and -2λ .

3 Telegraph equation and the gap atlas

For $c > 0$, writing the total density $\rho = f_+ + f_-$ and flux $j = f_+ - f_-$, the forward (adjoint) equations give $\rho_t = -cj_x$ and $j_t = -c\rho_x - 2\lambda j$. Elimination yields

$$\rho_{tt} + 2\lambda\rho_t = c^2\rho_{xx}. \quad (3)$$

For $c > 0$ and $\lambda > 0$, the sharp spectral-abscissa gap from the invariant constant mode is

$$g(c, \lambda) = \begin{cases} \lambda, & \lambda \leq c, \\ \lambda - \sqrt{\lambda^2 - c^2}, & \lambda > c. \end{cases}$$

It is zero when $c = 0$ or $\lambda = 0$. The word “gap” here means spectral abscissa: it is not a constant-free L^2 operator-norm decay assertion. Non-normal and critical blocks can have polynomial transient factors.

4 Stationary and essential boundaries

For $c > 0, \lambda > 0$, only the constant density is stationary. If $c = 0$, positions are frozen and every common spatial profile is stationary, an infinite-dimensional space. If $\lambda = 0$ and $c > 0$, the stationary space is exactly the two velocity-wise spatial constants. At $c = \lambda = 0$ all functions are stationary.

For $c > 0, \lambda > 0$, high-frequency blocks satisfy (this is an essential quotient statement, not a finite-mode estimate)

$$\lim_{|k| \rightarrow \infty} \|e^{tG_k}\| = e^{-\lambda t}$$

in the essential quotient on the complement of constants. Consequently the essential norm is $e^{-\lambda t} > 0$ and the semigroup is noncompact; hence it belongs to no finite Schatten class. On $c = 0$ or $\lambda = 0$ the essential norm is one. The latter face is the same-clock unitary pair of translations, whereas $c = 0$ is velocity-only mixing without spatial decay.

5 Receipt and scope boundary

The receipt contains $5 \times 5 \times 7 \times 4 = 700$ exact/high-precision block rows and 25 gap rows. A producer-independent checker reconstructs every matrix entry and eigenvalue at 100 decimal working digits; every nonzero serialized decimal has 82 significant digits. SymPy checks (1), the characteristic polynomial, (3), the critical condition, and the diffusive gap identity. Clean replay is byte-identical and hostile mutations include repaired-hash, unknown-key and stale-hash attacks.

The finite grid does not carry the all-mode quantifiers; those follow from the displayed identities and the $|k| \rightarrow \infty$ argument. The finite Fourier characteristic polynomial is source-local, not a Fredholm determinant or a target divisor. No arithmetic local data, Euler factors, root numbers, automorphy, Hilbert–Polya operator, or Route-B invocation is claimed.

Route-A decision. The locked tuple is (A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_FORMAL_HINT), with overall ROUTE_A_REJECTED; Route B is false. The historical telegraph framework is cited to Kac (1974), Rocky Mountain Journal of Mathematics 4(3), 497–509. The scope literal is NO_BAD_EULER_OR_ROOT_NUMBER.